



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 7 · STANDARD 6.AT.8

Write Equations

Flagship

Lesson 7-1-flagship · Teacher Copy

Name: _____

Date: _____

Class: _____

COLD CASE MISSION

The Equation Vault

You are Detective Ruiz's new code analyst. A tip just came in: 'A number of stolen gems plus 8 more equals 20 total.' The evidence vault only opens when the clue is translated into a correct equation. Master writing equations and the case cracks wide open.



TODAY'S OBJECTIVES

Content Objective: I can write an equation to represent a real-world situation.

Language Objective: I can explain my equation using the words equation, variable, equal sign, and expression.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Equation

Variable

Equal sign

Expression



Tap any word to see what it means and a picture.

- 1 A math sentence that says two amounts are equal, with an = sign, is an

- 2 A letter that stands for an unknown number is a .

- 3 The symbol = that shows two amounts are the same is the .

- 4 A group of numbers and variables with no equal sign is an .

Reflect — Exit Ticket

Which equation represents: 'A number divided by 6 equals 9'?

A. $n / 6 = 9$

B. $6n = 9$

C. $n - 6 = 9$

D. $n + 6 = 9$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Equation
2. **Notes 2:** Variable
3. **Notes 3:** Equal sign
4. **Notes 4:** Expression
5. **Watch for this mistake:** *A common mistake in Write Equations is skipping the key idea: "Find the unknown, choose a variable, then turn the words into symbols around an equal sign." — always check your work against this rule before you submit.*
6. **Try It 1:** A. $6c = 54$ — *Let c be the number of stolen coins. 'Six times' means multiply by 6, so the equation is $6c = 54$.*
7. **Try It 2:** A. $b / 5 = 4$ — *Let b be the total gold bars. Sharing equally into 5 cases means $b / 5$, and each case has 4, so $b / 5 = 4$.*
8. **Exit Ticket:** A. $n / 6 = 9$ — *'Divided by 6' means division: $n / 6 = 9$. Check: $54 / 6 = 9$ ✓*